

A Mathematical Framework for Psychophysical Interaction: Discrete Foundations, Continuous Extension, and Behavioral Applications

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Abstract

We develop a unified mathematical framework for interpersonal interaction grounded in psychophysical perception laws, with particular emphasis on Weber–Fechner scaling. By explicitly distinguishing between active (expressive) and passive (perceptual) interaction variables, we formulate a discrete-time linear dynamical system governed by an *interaction resistivity matrix* γ . We provide a complete analytical characterization of this discrete system, including closed-form eigenvalue spectra, necessary and sufficient stability conditions via the Jury criterion, and spectral bifurcation regimes corresponding to convergence, oscillation, and divergence of interaction trajectories. We then develop a continuous-time extension by introducing the *interaction flow matrix* M as the infinitesimal generator, and prove the generator-matrix equivalence $\gamma = \exp(M\Delta t)$, establishing formal algebraic consistency between discrete and continuous systems. The continuous formulation enables rigorous characterization of interaction trajectories using spectral theory, stability analysis, and Hopf-type bifurcation conditions. We further demonstrate that classical behavioral models—including Thorndike’s learning curves and Gottman’s dyadic marital interaction equations—emerge naturally as discrete or reduced-order representations of the continuous interaction flow. The framework does not claim psychological universality; rather, it establishes a mathematically coherent foundation generating falsifiable predictions and clarifying how perception–action coupling alone produces rich interaction dynamics.

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1 Introduction

1.1 Motivation and Context

Interpersonal interaction represents a fundamental class of dynamical systems in psychological and social sciences. Despite extensive empirical work, a unified mathematical framework connecting psychophysical principles to interaction dynamics remains elusive. This paper addresses this gap by constructing a theoretical model grounded in the Weber–Fechner law—a cornerstone of psychophysics relating stimulus intensity to perceptual response [2].

Human interaction requires the continuous transformation of perceived input into expressive output. Any mathematical treatment of interaction must therefore begin with how perception itself behaves as a function of stimulus. Psychophysics provides a formal answer: perception scales nonlinearly. The Weber–Fechner law, established for physical sensations such as brightness, weight, and sound, has more recently found support in emotional and social domains. Berkovich et al. [1] demonstrated that pleasant emotional responses exhibit logarithmic scaling with stimulus intensity. Research by Kahneman and Tversky [5] in Prospect Theory supported analogous scaling in psychological utility. Neurobiological studies by Namboodiri et al. [7] and Lewis [6] confirmed that brain regions for reward perception—specifically the ventral striatum and orbitofrontal cortex—operate in accordance with Weber’s law. These converging lines of evidence suggest that interpersonal interaction may itself obey measurable quantitative laws.

1.2 Scope and Intent

This work is theoretical in nature. Its purpose is not to propose a complete empirical model of interpersonal behavior, but to develop a mathematically coherent framework within which discrete interaction models and continuous-time dynamics can be analyzed under a shared formal structure. We assume that social-emotional stimuli obey logarithmic perceptual scaling within moderate-intensity domains as a local approximation, not as a universal psychological law. Extreme affective states, saturation effects, or nonlinear valuation regimes are beyond the present scope and may require alternative psychophysical models (e.g., Stevens’ power law). Empirical validation and parameter estimation are treated as future directions.

1.3 Contributions

This paper provides six principal advances:

1. We introduce a formal mathematical model embedding psychophysical perception directly into interaction dynamics via the *interaction resistivity matrix* γ .

2. We provide a complete analytical characterization of the resulting discrete-time system, including eigenvalue spectra, Jury stability conditions, and bifurcation regimes.
3. We derive the *continuous-time active interaction equation*

$$\frac{dI_S}{dt} = MI_S(t),$$

establishing continuous-time psychophysical dynamics from first principles.

4. We prove the *generator equivalence theorem* $\gamma = e^{M\Delta t}$, ensuring algebraic consistency between discrete and continuous systems.
5. We characterize stability and phase transitions using eigenvalue analysis and spectral theory, identifying thresholds that separate convergence, escalation, oscillation, and neutral equilibrium regimes.
6. We demonstrate applicability by recovering Thorndike’s learning curve and showing that the Gottman marital interaction model is a discrete projection of the continuous theory.

1.4 Paper Organization

Section 2 reviews psychophysical foundations. Section 3 introduces core mathematical objects. Sections 4 and 5 develop discrete and continuous dynamics, respectively. Section 6 presents spectral theory and bifurcation analysis. Section 7 unifies the framework with the Gottman model. Section 8 presents behavioral applications. Section 9 discusses implications, and Section 10 concludes.

2 Psychophysical Foundations

2.1 The Weber–Fechner Law

The Weber–Fechner law [2] states that perceived intensity P responds logarithmically to a physical stimulus magnitude S :

$$P = \alpha \ln\left(\frac{S}{S_0}\right), \tag{1}$$

where

- $P \in \mathbb{R}$ denotes perceptual intensity (psychophysical units),
- $S \in \mathbb{R}^+$ denotes physical stimulus magnitude,

- $S_0 \in \mathbb{R}^+$ is a baseline threshold at which perception is zero,
- $\alpha \in \mathbb{R}^+$ is a perceptual sensitivity constant.

Equation (1) follows from Weber’s differential law,

$$dP = \alpha \frac{dS}{S}, \quad (2)$$

which integrates to the logarithmic form above. This formulation implies *diminishing marginal perceptual returns*—a doubling of stimulus intensity yields only an additive increase in subjective perception.

2.2 Extension to Social-Emotional Stimuli

Assumption 2.1 (Restricted Psychophysical Generalization). *In this work, we assume that social-emotional stimuli obey logarithmic perceptual scaling within moderate-intensity regimes:*

$$P_{\text{social}} \approx \alpha \ln \left(\frac{S_{\text{social}}}{S_0} \right),$$

where α is interpreted as perceptual affective sensitivity. This assumption is adopted as a local approximation rather than a universal psychological law.

This assumption rests on three independent pillars:

1. **Neurobiological Evidence.** Berkovich et al. [1] demonstrated that pleasant emotional responses exhibit logarithmic scaling with stimulus intensity, with effect sizes comparable to classical psychophysical modalities. The ventral striatum and orbitofrontal cortex demonstrate logarithmic encoding of reward magnitude, consistent with Weber–Fechner mechanisms.
2. **Decision-Theoretic Support.** Prospect Theory [5] and neuroscience studies by Namboodiri et al. [7] confirm that subjective value perception follows Weber’s law. Lewis [6] further supports this through dynamic systems modeling of emotional and neurobiological coupling.
3. **Theoretical Parsimony.** Social interaction fundamentally involves perception-response coupling. Assuming the same psychophysical mechanisms govern social stimuli provides a unified framework without introducing ad hoc assumptions.

Extreme stimuli—trauma, pathological affect, or saturation—may require Stevens’ power law or other nonlinear models, which are beyond the present scope.

2.3 Perception-Action Feedback as Mathematical Structure

Human behavior does not stop at perception. Internal states produce expression. A complete model therefore requires two maps:

$$S \longleftrightarrow I_S \longleftrightarrow I_P \longleftrightarrow P,$$

stimulus $\leftrightarrow$ active interaction $\leftrightarrow$ passive interaction $\leftrightarrow$ perception. The central mathematical hypothesis of this paper is that the mapping between perception and expression is linear in its encoded domain—a structure that gives rise to the interaction resistivity matrix.

3 Mathematical Objects and Core Definitions

3.1 Interaction Variables

Definition 3.1 (Active Interaction — Expressive Signal). *For a system of n interacting agents, the active interaction vector is*

$$I_S(t) \in \mathbb{R}^n,$$

whose component $[I_S(t)]_i$ represents the expressive interaction signal generated by agent i at time t .

Definition 3.2 (Passive Interaction — Perceptual Reception). *The passive interaction vector is*

$$I_P(t) \in \mathbb{R}^n,$$

where $[I_P(t)]_j$ denotes the perceptual value experienced by agent j at time t . Passive interaction is an internal perception state that mediates how external input is internally encoded.

3.2 Encoding and Decoding Maps

Let $S_i(t)$ denote the physical or social stimulus received by agent i . Using Weber–Fechner scaling, the passive (perceptual) encoding is:

$$[I_P(t)]_i = k_{P,i} \alpha_i \ln \left(\frac{S_i(t)}{S_0} \right), \quad (3)$$

and the active (expressive) encoding is:

$$[I_S(t)]_i = k_{S,i} \ln \left(\frac{S_i(t)}{S_0} \right), \quad (4)$$

where $k_{P,i}$ is a perceptual interpretation constant and $k_{S,i}$ is an expressive sensitivity coefficient.

3.3 The Interaction Resistivity Matrix

Theorem 3.1 (Linear Perception–Expression Mapping). *There exists a constant matrix $\gamma \in \mathbb{R}^{n \times n}$, called the interaction resistivity matrix, such that*

$$I_P(t) = \gamma I_S(t), \quad (5)$$

where $\gamma_{ij} = \frac{k_{P,i} \alpha_i}{k_{S,j}}$.

Proof. From equations (3) and (4),

$$\frac{[I_P(t)]_i}{[I_S(t)]_j} = \frac{k_{P,i} \alpha_i}{k_{S,j}}.$$

Define each entry $\gamma_{ij} := \frac{k_{P,i} \alpha_i}{k_{S,j}}$. Summing influence from all relevant perceptual inputs:

$$[I_P(t)]_i = \sum_{j=1}^n \gamma_{ij} [I_S(t)]_j,$$

which gives (5) in vector form. □

Remark 3.1. *The scalar version of this relation is $I_P = \gamma I_S$ with $\gamma = k_P \alpha / k_S$, equivalently written as $I_S = \beta I_P$ where $\beta = k_S / (k_P \alpha) = 1/\gamma$ is the interaction permeability coefficient.*

The diagonal elements γ_{ii} represent self-perception feedback loops, capturing how an agent’s own perceived state influences its subsequent expression. Off-diagonal elements γ_{ij} encode cross-agent influence, reflecting how perception of another agent modulates one’s response.

4 Discrete-Time Interaction Dynamics

4.1 Discrete Interaction System

Interpersonal interaction unfolds over time through repeated perception–action cycles. We represent this evolution using a discrete-time recurrence relation, suitable for conversational turn-taking or sampled emotional states.

Definition 4.1 (Discrete Interaction Dynamics). *Let $I_S(t) \in \mathbb{R}^n$ denote the active interaction state vector at time step t . Under linear perception-expression transformation at local equilibrium, the system evolves according to the linear recurrence*

$$I_S(t+1) = \gamma I_S(t), \quad t \in \mathbb{N}. \quad (6)$$

Remark 4.1. *The discrete-time formulation represents interaction sampled at regular intervals or conversational turns. The linear recurrence does not imply that interaction itself is inherently discrete, but rather that discrete observations admit a linear update representation.*

4.2 Existence and Uniqueness of Solutions

Theorem 4.1 (Existence and Uniqueness of Discrete Evolution). *For any initial state $I_S(0) \in \mathbb{R}^n$, the recurrence (6) admits a unique solution given by*

$$I_S(t) = \gamma^t I_S(0), \quad t \in \mathbb{N}. \quad (7)$$

Proof. By induction: assume $I_S(t) = \gamma^t I_S(0)$. Then $I_S(t+1) = \gamma I_S(t) = \gamma(\gamma^t I_S(0)) = \gamma^{t+1} I_S(0)$. $\square$

4.3 Equilibrium and Fixed-Point Structure

A state I^* is an equilibrium if $I^* = \gamma I^*$, i.e., $(\gamma - \mathbf{I})I^* = 0$.

Lemma 4.2. *The equilibrium set is the nullspace of $(\gamma - \mathbf{I})$. The equilibrium is unique (i.e., $I_S^* = 0$) if and only if $\gamma - \mathbf{I}$ is full rank.*

4.4 Time-Varying and Adaptive Interactions

The framework naturally extends to adaptive or context-dependent interactions. Let $\gamma(t)$ denote a time-varying resistivity matrix:

$$I_S(t+1) = \gamma(t)I_S(t), \quad I_S(t) = \left(\prod_{i=0}^{t-1} \gamma(i) \right) I_S(0).$$

This generalization accommodates learning, fatigue, habituation, or strategic adaptation.

4.5 Stability Analysis: Spectral Radius Criterion

The long-term behavior of system (6) is governed by the *spectral radius*:

$$\rho(\gamma) := \max_i |\lambda_i(\gamma)|.$$

Proposition 4.3 (Asymptotic Stability Criterion). *The equilibrium $I_S = 0$ is asymptotically stable if and only if $\rho(\gamma) < 1$.*

Proof. Let $\gamma = V\Lambda V^{-1}$ be a Jordan decomposition over $\mathbb{C}$. Then $I_S(t) = V\Lambda^t V^{-1} I_S(0)$, and each Jordan block grows as $O(t^k |\lambda|^t)$. Hence $I_S(t) \rightarrow 0$ if and only if $|\lambda_i| < 1$ for all i . $\square$

4.6 Dyadic-Agent Specialization

For the dyadic case $n = 2$:

$$I_S(t+1) = \begin{pmatrix} \gamma_{AA} & \gamma_{AB} \\ \gamma_{BA} & \gamma_{BB} \end{pmatrix} I_S(t).$$

Definition 4.2 (Characteristic Polynomial and Eigenvalues). *The characteristic polynomial is*

$$p(\lambda) = \lambda^2 - \text{tr}(\gamma)\lambda + \det(\gamma),$$

with closed-form eigenvalues

$$\lambda_{\pm} = \frac{\text{tr}(\gamma) \pm \sqrt{\text{tr}(\gamma)^2 - 4\det(\gamma)}}{2}. \quad (8)$$

Theorem 4.4 (Rigorous Stability via Jury Criterion). *The equilibrium $I_S = 0$ is asymptotically stable if and only if:*

- (i) $|\det(\gamma)| < 1$,
- (ii) $1 + \text{tr}(\gamma) + \det(\gamma) > 0$,
- (iii) $1 - \text{tr}(\gamma) + \det(\gamma) > 0$.

Proof. The system is stable if and only if both roots of $\lambda^2 - \text{tr}(\gamma)\lambda + \det(\gamma) = 0$ lie strictly inside the unit disk. Setting $a_1 = -\text{tr}(\gamma)$ and $a_0 = \det(\gamma)$, the Jury stability criterion for second-order polynomials requires $|a_0| < 1$, $1 + a_1 + a_0 > 0$, and $1 - a_1 + a_0 > 0$, yielding conditions (i)–(iii). $\square$

Corollary 4.5 (Compact Stability Region). *If $\det(\gamma) \geq 0$, stability is equivalent to $|\text{tr}(\gamma)| < 1 + \det(\gamma) < 2$.*

4.7 Discriminant and Oscillatory Regimes

Definition 4.3 (Discriminant and Conflict Parameter). *Define the discriminant $\Delta = \text{tr}(\gamma)^2 - 4\det(\gamma)$ and the conflict parameter $C = -\gamma_{AB}\gamma_{BA}$. Then $\Delta = (\gamma_{AA} - \gamma_{BB})^2 - 4C$.*

Theorem 4.6 (Oscillation Criterion for Discrete System). *The discrete system exhibits oscillatory dynamics if and only if*

$$C > \frac{(\gamma_{AA} - \gamma_{BB})^2}{4}. \quad (9)$$

Proof. Oscillations occur when eigenvalues are complex, i.e., $\Delta < 0$. Substituting the expression for Δ yields the stated inequality. $\square$

4.8 Bifurcation Theory for the Discrete System

In discrete-time systems, qualitative changes occur when eigenvalues of $\gamma(\mu)$ cross the unit circle in the complex plane.

Definition 4.4 (Spectral Bifurcation Point). *A parameter value $\mu = \mu^*$ is a spectral bifurcation point if $\rho(\gamma(\mu^*)) = 1$ and the qualitative nature of the eigenvalues changes as μ crosses μ^* .*

Proposition 4.7 (Eigenvalue Phase Transition). *Let $\gamma(\mu) = \begin{pmatrix} \gamma_{AA} & \mu \\ -\mu & \gamma_{BB} \end{pmatrix}$ with fixed $\gamma_{AA}, \gamma_{BB} \in \mathbb{R}$. The eigenvalues are real for $|\mu| < \mu^*$ and form a complex conjugate pair for $|\mu| > \mu^*$, where $\mu^* = |\gamma_{AA} - \gamma_{BB}|/2$.*

A Hopf-like (Neimark–Sacker-type) transition occurs at $\mu = \mu_c$ if $\gamma(\mu_c)$ has a complex conjugate eigenvalue pair $\lambda_{\pm} = e^{\pm i\theta}$ and $\frac{d}{d\mu}|\lambda_{\pm}(\mu_c)| \neq 0$. Under these conditions, the linear system transitions from monotonic decay to oscillatory behavior. This transition may correspond to the emergence of cyclical emotional patterns, escalation–de-escalation loops, or resonance between agents—arising as a purely structural consequence of perception–action coupling.

5 Continuous-Time Extension and the Interaction Flow Matrix

5.1 Motivation for Continuous-Time Dynamics

The discrete system (6) models interaction as if change occurs only at discrete emotional “turns.” To represent real-time affective evolution, we let the time step shrink and derive a continuous-time system.

Let $\gamma(\Delta t)$ denote the resistivity matrix for time increment Δt . Then:

$$I_S(t + \Delta t) = \gamma(\Delta t)I_S(t),$$

and subtracting $I_S(t)$:

$$\frac{\Delta I_S(t)}{\Delta t} = \frac{\gamma(\Delta t) - \mathbf{I}}{\Delta t} I_S(t).$$

Taking the limit $\Delta t \rightarrow 0$ yields a differential equation of the form $\dot{I}_S = M I_S$.

5.2 Definition of the Interaction Flow Matrix

Definition 5.1 (Interaction Flow Matrix). *We restrict attention to families $\gamma(\Delta t)$ that are differentiable at $\Delta t = 0$ and generate bounded linear operators on $\mathbb{R}^n$. The interaction flow matrix (generator) is*

$$M := \lim_{\Delta t \rightarrow 0} \frac{\gamma(\Delta t) - \mathbf{I}}{\Delta t}, \quad (10)$$

provided the limit exists. This parallels generator theory in semigroup analysis for linear dynamical systems.

5.3 Existence and Uniqueness of Continuous Evolution

Theorem 5.1 (Existence and Uniqueness of Continuous Dynamics). *If M is bounded, then the system*

$$\frac{dI_S}{dt} = M I_S(t) \quad (11)$$

admits a unique solution:

$$I_S(t) = e^{Mt} I_S(0), \quad e^{Mt} := \sum_{k=0}^{\infty} \frac{(Mt)^k}{k!}. \quad (12)$$

Proof. Standard Picard–Lindelöf theorem for linear ODEs with constant coefficients. $\square$

5.4 Generator-Matrix Equivalence Theorem

The following theorem establishes the mathematical bridge between discrete and continuous interaction systems.

Theorem 5.2 (Generator-Matrix Equivalence). *Let M be the interaction flow generator. Then for any step size $\Delta t > 0$:*

$$\gamma(\Delta t) = e^{M\Delta t}, \quad (13)$$

and conversely, if γ is invertible:

$$M = \frac{1}{\Delta t} \log(\gamma). \quad (14)$$

Proof. Iterating the continuous solution (11) over $t = k\Delta t$:

$$I_S(k\Delta t) = e^{Mk\Delta t} I_S(0) = (e^{M\Delta t})^k I_S(0).$$

From the discrete evolution, $I_S(k\Delta t) = \gamma^k I_S(0)$. Equating both expressions gives $\gamma = e^{M\Delta t}$. If γ is nonsingular, the matrix logarithm exists and yields $M = \frac{1}{\Delta t} \log(\gamma)$. $\square$

5.5 Spectral Interpretation

Let $\lambda_i(\gamma)$ be an eigenvalue of γ . From (13):

$$\lambda_i(\gamma) = e^{\lambda_i(M)\Delta t} \implies \lambda_i(M) = \frac{1}{\Delta t} \log(\lambda_i(\gamma)).$$

Thus:

- $|\lambda_i(\gamma)| < 1 \implies \text{Re}(\lambda_i(M)) < 0$ (stable),
- $|\lambda_i(\gamma)| > 1 \implies \text{Re}(\lambda_i(M)) > 0$ (unstable),
- $\text{Im}(\lambda_i(M)) \neq 0$ produces oscillatory trajectories.

6 Spectral Theory, Oscillation, and Qualitative Regime Change

6.1 Continuous-Time Stability Criterion

Theorem 6.1 (Continuous-Time Stability Criterion). *The equilibrium $I_S = 0$ of $\dot{I}_S = MI_S$ is asymptotically stable if and only if*

$$\text{Re}(\lambda_i(M)) < 0 \quad \text{for all } i. \quad (15)$$

Proof. Since $I_S(t) = e^{Mt} I_S(0)$, diagonalizing yields $I_S(t) = V \text{diag}(e^{\lambda_i(M)t}) V^{-1} I_S(0)$. The term $e^{\lambda_i(M)t}$ contracts to 0 if and only if $\text{Re}(\lambda_i(M)) < 0$. $\square$

6.2 Two-Agent Spectral Geometry

For $n = 2$, with $M = \begin{pmatrix} m_{AA} & m_{AB} \\ m_{BA} & m_{BB} \end{pmatrix}$, define the *continuous conflict parameter*:

$$C := -m_{AB}m_{BA}, \quad \Delta = \text{tr}(M)^2 - 4 \det(M) = (m_{AA} - m_{BB})^2 - 4C.$$

Theorem 6.2 (Oscillatory Interaction Criterion — Continuous System). *The continuous system exhibits oscillatory trajectories if and only if*

$$\Delta < 0 \iff C > \frac{(m_{AA} - m_{BB})^2}{4}. \quad (16)$$

Proof. Oscillations arise when eigenvalues of M are complex, i.e., $\lambda_{\pm} = \alpha \pm i\beta$ with $\beta \neq 0$, which holds if and only if $\Delta < 0$. Substituting the expression for Δ yields the stated inequality. $\square$

6.3 Qualitative Interaction Phase Regimes

Definition 6.1 (Interaction Phase Regimes). *Let $\lambda = \lambda_i(M)$. The system admits the following qualitative regimes:*

<i>Regime</i>	<i>Condition on λ</i>	<i>Interpretation</i>
<i>Convergence / De-escalation</i>	$\text{Re}(\lambda) < 0, \text{Im}(\lambda) = 0$	<i>Monotonic emotional calming</i>
<i>Divergence / Escalation</i>	$\text{Re}(\lambda) > 0$	<i>Spiraling conflict, runaway affect</i>
<i>Neutral equilibrium</i>	$\text{Re}(\lambda) = 0, \text{Im}(\lambda) = 0$	<i>Emotional inertia</i>
<i>Oscillatory cycling</i>	$\text{Im}(\lambda) \neq 0$	<i>Cyclical emotional exchange</i>
<i>Oscillatory instability</i>	$\text{Re}(\lambda) > 0, \text{Im}(\lambda) \neq 0$	<i>Escalating fight–recovery loops</i>

6.4 Hopf-Type Bifurcation in Continuous Time

Let $M = M(\mu)$ depend smoothly on a parameter μ .

Definition 6.2 (Continuous Hopf-like Transition). *A bifurcation occurs at $\mu = \mu_c$ if:*

1. $M(\mu_c)$ has a conjugate pair $\lambda_{\pm} = \pm i\omega$,
2. $\frac{d}{d\mu} \text{Re}(\lambda_{\pm}(\mu_c)) \neq 0$.

Proposition 6.3. *Near μ_c , trajectories transition from monotonic relaxation to oscillatory relational cycling:*

$$I_S(t) \sim e^{\alpha t} \longrightarrow I_S(t) \sim e^{\alpha t} \cos(\omega t),$$

where $\alpha = \text{Re}(\lambda_{\pm})$.

6.5 Interpretation for Social Interaction Science

- If mutual influence (m_{AB}, m_{BA}) grows large relative to self-regulation asymmetry, cycles emerge. *Psychological interpretation:* people emotionally “resonate” more than they self-contain.
- If self-regulation (m_{AA}, m_{BB}) dominates, interaction decays—characteristic of resilient, calm dyads.
- If $\text{Re}(\lambda) > 0$, affect escalates inherently, corresponding to toxic, unstable relationships independent of external triggers.

7 Unification with the Gottman Marital Interaction Model

7.1 The Gottman Model

Gottman et al. [3] proposed a dyadic interaction model:

$$\begin{aligned} W(t+1) &= a + r_1 W(t) + I_{HW}(H(t)), \\ H(t+1) &= b + r_2 H(t) + I_{WH}(W(t)), \end{aligned} \quad (17)$$

where $W(t)$ and $H(t)$ denote the emotional states of two partners, r_1, r_2 represent emotional inertia, and I_{HW}, I_{WH} are influence functions.

Under local linearization around equilibrium:

$$I_{HW}(H) = c_{HW}H, \quad I_{WH}(W) = c_{WH}W,$$

the homogeneous dynamics (after equilibrium shift $y(t) = x(t) - x^*$) reduce to:

$$y(t+1) = \begin{pmatrix} r_1 & c_{HW} \\ c_{WH} & r_2 \end{pmatrix} y(t). \quad (18)$$

7.2 Identification with the Interaction Resistivity Framework

Proposition 7.1 (Parameter Identification). *The linearized Gottman model is a special case of the interaction resistivity framework under:*

$$\gamma_{AA} = r_1, \quad \gamma_{BB} = r_2, \quad \gamma_{AB} = c_{HW}, \quad \gamma_{BA} = c_{WH}.$$

Theorem 7.2 (Stability Equivalence). *The Gottman model is asymptotically stable about its equilibrium if and only if $\rho(\gamma) < 1$, where γ is the corresponding interaction resistivity matrix.*

Proof. The stability of the Gottman dynamics reduces to $\rho(\gamma) < 1$ by Theorem 4.4 and Proposition ??, since both systems share the same governing matrix. $\square$

Proposition 7.3. *Gottman's empirical stability condition $|r_1 r_2 - c_{HW} c_{WH}| < r_1 r_2$ is equivalent to the Jury criterion applied to γ , since $\det(\gamma) = r_1 r_2 - c_{HW} c_{WH}$.*

7.3 Identification with the Continuous Flow Generator

By Theorem 5.2:

$$\gamma(\Delta t) = e^{M\Delta t} \implies M = \frac{1}{\Delta t} \log(\gamma).$$

Thus the Gottman model is the discrete projection of the continuous psychophysical flow model. The psychological meanings are summarized in Table 1.

Table 1: Psychological interpretation of Gottman–framework correspondence.

Parameter	Psychological meaning
r_1, r_2 (diagonal of γ)	Emotional inertia (self-feedback resistivity)
c_{WH}, c_{HW} (off-diagonal of γ)	Cross-coupling influence functions
Sign of $\text{Re}(\lambda(M))$	Emotional stability / escalation
$\text{Im}(\lambda(M)) \neq 0$	Oscillatory conflict cycles

8 Applications and Case Studies

The following examples are intended as illustrative embeddings rather than empirical tests of the proposed framework. Our objective is not merely to reinterpret these models, but to show that they mathematically emerge as consequences of the continuous interaction equation.

8.1 Thorndike Puzzle-Box Learning Curve

Thorndike [8] observed that trial-based escape performance decays monotonically: escape time $t_S(n)$ decreases with the number of attempts n . We interpret escape performance as an active interaction magnitude, modulated by learning across trials.

8.1.1 Mathematical Setup

Let $t_S(n)$ be the escape time after n trials, $I_S(n)$ the active interaction performance magnitude, and $A > 0$ a learning sensitivity constant. The first-order learning decay assumption gives:

$$\frac{dI_S}{dt} = -AI_S. \quad (19)$$

8.1.2 Linking Learning to Interaction Flow

The continuous interaction theory states:

$$\frac{dI_S}{dt_S} = -M(t_S)I_S.$$

Applying the chain rule:

$$\frac{dI_S}{dn} = \frac{dI_S}{dt_S} \cdot \frac{dt_S}{dn} = M(t_S)I_S \cdot \frac{dt_S}{dn}.$$

Equating with (19) and dividing by $I_S \neq 0$:

$$\frac{dt_S}{dn} = -AM^{-1}(t_S). \quad (20)$$

Equation (20) is the canonical continuous-interaction interpretation of Thorndike’s learning curve: escape time decreases at a rate inversely proportional to system flow.

8.1.3 Closed-Form Example

Suppose the (scalar) flow satisfies $\|M(t_S)\| = c/t_S$ for constant $c > 0$ (empirically consistent with diminishing learning rate). Then from (20):

$$\frac{dt_S}{dn} = -\frac{A}{c} t_S \implies \frac{dt_S}{t_S} = -\frac{A}{c} dn.$$

Solving by separation of variables:

$$t_S(n) = t_S(0) \exp\left(-\frac{A}{c} n\right). \quad (21)$$

This replicates Thorndike’s exponential decay curve mathematically rather than descriptively. Figure 1 shows the solution with parameters $A = 5$, $c = 5$, and $t_S(0) = 180$ seconds.

8.2 Summary of Application Results

Table 2: Mathematical explanations of behavioral phenomena.

Phenomenon	Mathematical explanation
Thorndike learning decay	Emerges from inverse flow matrix
Emotional resonance	High cross-coupling $\Rightarrow$ complex eigenvalues
Relationship escalation	Positive eigenvalue real part
Conflict cycles	Imaginary eigenvalue component

9 Discussion

The mathematical framework developed in this paper reframes interpersonal interaction as a flow system governed by linear operators. Rather than treating emotional change as qualitative or phenomenological, the model asserts that patterns of expression and perception can be described using state variables and semigroup theory. Three conceptual implications emerge:

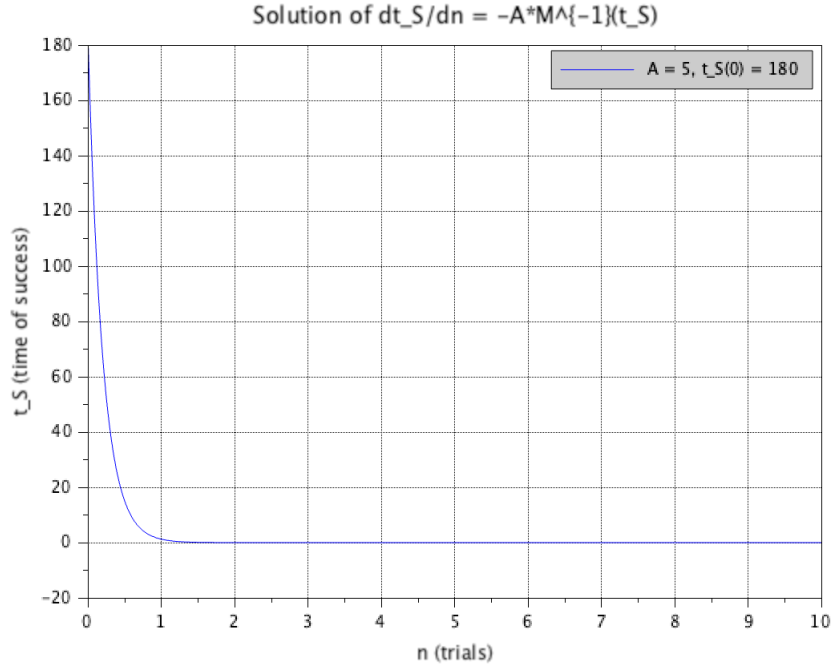


Figure 1: Escape time t_S versus trial number n from (21) with $M(t_S) = 5/t_S$, $A = 5$, $t_S(0) = 180$ s. This reproduces the “time-curve” graph of Thorndike’s puzzle-box experiments [8].

1. **Interaction is a dynamical process, not a static record.** Where traditional psychology models emotions as snapshots, the equation $\dot{I}_S = M I_S$ asserts that affect evolves continuously—like motion in physics—governed by intrinsic rates of change. This shifts the epistemology of emotional science from measurement to mechanism.
2. **Relationship behavior is a spectral phenomenon.** The eigenvalues of the flow matrix M determine convergence, escalation, or oscillation. Thus emotional outcomes are not merely products of personality or context, but consequences of structural parameters (self-regulation, cross-influence). This parallels physics: just as vibration requires specific mass and tension ratios, cyclical emotional patterns arise when cross-agent influence exceeds stability thresholds.
3. **Classical behavioral phenomena emerge as mathematical corollaries.** Thorndike’s learning curve and Gottman’s dyadic interaction model are shown not as separate theories, but as limiting cases of the same underlying dynamical principle, unifying learning theory and relationship science under a shared mathematical umbrella.

Several clarifications are essential. First, this model does not claim to explain internal emotional experience, subjective meaning, or narrative cognition. Interaction variables

are abstract quantities representing measurable interactional influence, not phenomenological states. Second, oscillatory or divergent dynamics derived here should not be interpreted as pathological or maladaptive per se; they arise as mathematical consequences of linear feedback coupling. Third, the equivalence with the Gottman model is structural and dynamical, not epistemological: the present framework does not replace empirical marital research, nor does it reinterpret Gottman’s constructs as psychophysical variables. Different choices of psychophysical scaling (e.g., power-law) would lead to modified but structurally analogous generators.

10 Conclusion

This paper introduced a unified theoretical-mathematical structure for modeling interpersonal interaction. Beginning from Weber–Fechner sensory encoding, we constructed a two-layer representation of affect—active expression and passive perception—linked by a resistivity matrix γ . We then formalized time evolution through:

$$I_S(t+1) = \gamma I_S(t), \quad \frac{dI_S}{dt} = M I_S, \quad \gamma = e^{M\Delta t}.$$

From this emerged a complete theory of interaction dynamics, including existence and uniqueness of solutions, spectral stability criteria (Jury conditions and real-part criteria), oscillation and bifurcation conditions, and the embedding of empirical social-behavioral models as special cases. The result is a mathematically rigorous account of affect dynamics—one that does not merely describe phenomena but derives them logically.

11 Future Work

Three principal avenues remain open:

1. **Nonlinear extension.** Real relationships include saturation, thresholding, and memory effects. Adding nonlinear terms $f(I_S)$ may reveal richer bifurcations.
2. **Stochastic interaction modeling.** Noise terms σdW_t allow modeling uncertainty and emotional volatility via stochastic differential equations.
3. **Parameter inference from empirical data.** A practical contribution would involve learning M or γ from observational dyadic recordings, yielding empirically calibrated interaction models.

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